

1. List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

IP Class	Range	Default Subnet Mask	Purpose	Real-world Example
Class A	1-126	255.0.0.0	Very large networks	A large MNC or ISP with millions of devices.
Class B	128-191	255.255.0.0	Medium to large networks	A large university campus or a major hospital network.
Class C	192-223	255.255.255.0	Small networks	A home Wi-Fi router or a small business office network
Class D	224-239	Not applicable	Multicast communication	IPTV streaming or live video conferencing.
Class E	240-255	Not applicable	Experimental and research purposes.	Networking research, protocol testing

2. Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses.

- Here, IP address is 192.168.10.45 and subnet mask is 255.255.255.0. First three octant of the IP address represents network portion and last octant is reserved for the hosts.
- If I set all host bits to 0 then Network address would be 192.168.10.0
- And if I set all host bits to 1 then Broadcast address would be 192.168.10.255

- The first address is the network address and last address is the broadcast address, so they cannot be assigned to hosts.
- And so, valid Host range would be between 192.168.10.1 to 192.168.10.254

3. Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

- Using the 10.0.0.0/24 (256 IP addresses), we can divide it into four equal-sized subnets. Splitting a /24 into four subnets gives /26 subnets, each with 64 IP addresses and 62 usable hosts.

Group	Network Address	Subnet Mask	CIDR	Usable Host Range	Broadcast Address
Delivery Partners	10.0.0.0	255.255.255.192	/26	10.0.0.1-10.0.0.62	10.0.0.63
Restaurants	10.0.0.64	255.255.255.192	/26	10.0.0.65-10.0.0.126	10.0.0.127
Customers	10.0.0.128	255.255.255.192	/26	10.0.0.129-10.0.0.190	10.0.0.191
Admins	10.0.0.192	255.255.255.192	/26	10.0.0.193-10.0.0.254	10.0.0.255

- Here, Base Network is: 10.0.0.0/24, Subnet mask is: 255.255.255.192(/26), Number of subnets are 4 and Usable hosts per subnet is 62.
- This subnetting plan provides a separate network as follows:
- Delivery Partners : 10.0.0.0/26
- Restaurants : 10.0.0.64/26
- Customers: 10.0.0.128/26
- Admins: 10.0.0.192/26

4. You are given the following scenario: A college Wi-Fi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet.

- In the given scenario, original network is: 172.16.0.0/16 and requirement is at least 20 subnets.
- Firstly, we need smallest number of bits in the calculation of 20 subnets.
- $2^4 = 16$ this is not enough.
- $2^5 = 32$ this is enough.
- So, 5 bits need to be borrowed from the host portion.
- Secondly, Original prefix is : /16 and borrowed subnets are : 5
- So, $16 + 5 = 21$
- Now, new subnet mask is : /21 in dotted decimal it is : 255.255.248.0
- Then, /21 network leaves : $32 - 21 = 11$ host bits
- Total addresses per subnet is now : $2^{11} = 2048$
- Here, Usable host addresses are : $2048 - 2 = 2046$
- In short, New subnet mask will be: 255.255.248.0/21, Number of host created is: 32 as compared to requirement of at least 20 usable hosts for each subnet is: 2046.

5. A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

- The error is that 192.168.1.300 is not a valid IPv4 address, each of the four octants must be between 0 and 255. Here, 300 is greater than 255, so this address is invalid and cannot be used.
- For the 192.168.1.x subnet, a valid IP address would look like: 192.168.1.10, 192.168.1.100 and 192.168.1.254
- With a typical 255.255.255.0 subnet mask, useable host addresses are generally between 192.168.1.1 to 192.168.1.254.
- Lastly, 192.168.1.0 is the network address and 192.168.1.255 is the broadcast address.